

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

If $9(4 - 3x) + 2 = 8(4 - 3x) + 18$, what is the value of $4 - 3x$?

Answer

- A. -16
- B. -4
- C. 4
- D. 16

Correct Answer: D

Rationale

Choice D is correct. The value of $4 - 3x$ can be found by isolating this expression in the given equation. Subtracting 2 from both sides of the given equation yields $9(4 - 3x) = 8(4 - 3x) + 16$. Subtracting $8(4 - 3x)$ from both sides of this equation yields $9(4 - 3x) - 8(4 - 3x) = 16$, which gives $1(4 - 3x) = 16$, or $4 - 3x = 16$. Therefore, the value of $4 - 3x$ is 16 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $4 - 3x$.

Choice C is incorrect and may result from conceptual or calculation errors.